General Mathematics — Exercise Sheet 2

Functions and the real numbers

Variant: Statements

Exercise 1 (Two finite-sum inductions). For $n \in \mathbb{N}$, the notation $\sum_{k=1}^{n} a_k$ means $a_1 + \cdots + a_n$. Prove by induction, using Lecture 2,

$$\sum_{k=1}^{n} k(k+1) = \frac{n(n+1)(n+2)}{3}, \qquad \sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

Exercise 2 (Negating function properties). Let $A \subseteq \mathbb{R}$ be nonempty and $f : A \to \mathbb{R}$. The function is bounded above if some $M \in \mathbb{R}$ satisfies $f(x) \le M$ for every $x \in A$. It is nondecreasing if for every $x, y \in A$, $x \le y$ implies $f(x) \le f(y)$. Write both definitions with quantifiers and negate them. Use the negations to show that $f(x) = x^2$ on $\mathbb{R}$ is neither bounded above nor nondecreasing. Use Lecture 1's negation rules.

Exercise 3 (A rational bijection). Let $A = \mathbb{R} \setminus \{-4\}$ and $B = \mathbb{R} \setminus \{2\}$. Prove that the function $f : A \to B$ given by

$$f(x) = \frac{2x - 3}{x + 4}$$

is well defined and bijective. Determine its inverse, including its domain and codomain. Determine exactly where $f(f(x))$ is defined and simplify it while preserving that domain. Use Lecture 2's inverse criterion.

Exercise 4 (Bounds that are attained and bounds that are not). Let $S = \{n/(n+1) : n \in \mathbb{N}\}$. Find its infimum and supremum. Decide whether each is a member of S. Prove leastness and greatestness using Lecture 2's Archimedean property, without assuming any sequence convergence theorem.

Exercise 5 (Hyperbolic functions and their inverse domains). Lecture 2 defines $\sinh x = (e^x - e^{-x})/2$ and $\cosh x = (e^x + e^{-x})/2$ for $x \in \mathbb{R}$. Define $\tanh x = \sinh x / \cosh x$ and $\operatorname{sech} x = 1/\cosh x$. Using only standing fact E's exponential/logarithm laws, prove that these quotients are defined everywhere and derive formulas and domains for the inverses of sinh on $\mathbb{R}$, cosh on $[0, \infty)$, tanh on $\mathbb{R}$, and sech on $[0, \infty)$. Also prove $1 - \tanh^2 x = \operatorname{sech}^2 x$.

Exercise 6 (Error bounds for a product and a quotient). Suppose $a, b \in \mathbb{R}$ and $0 < \eta < 1$ satisfy $|a - 1| \le \eta$ and $|b - 2| \le \eta$. Using Lecture 2's triangle and reverse-triangle inequalities, prove

$$|ab - 2| \le 3\eta + \eta^2, \qquad |a| \ge 1 - \eta > 0, \qquad \left|\frac{b}{a} - 2\right| \le \frac{3\eta}{1 - \eta}.$$